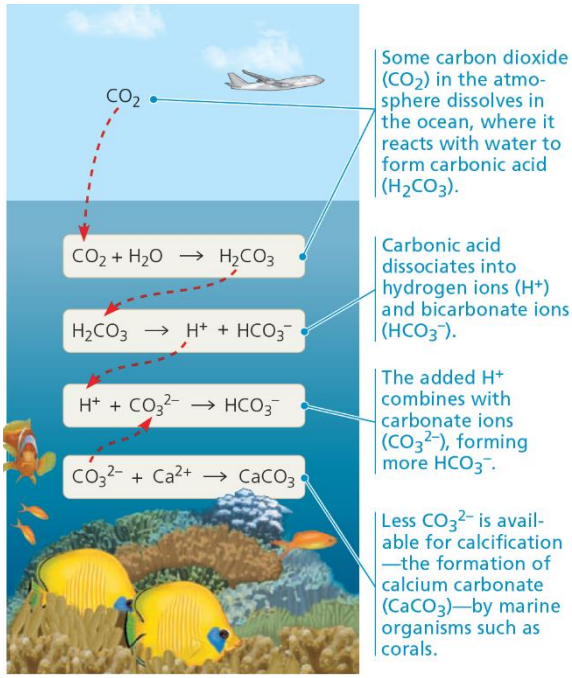
Question-0



▲ **Figure 3.11** **Atmospheric CO_2 from human activities and its fate in the ocean.**

$$\underset{\substack{H^+\text{ donor}\\(\text{acid})}}{H_2CO_3} \underset{\substack{\text{Response to}\\\text{a drop in pH}}}{\overset{\substack{\text{Response to}\\\text{a rise in pH}}}{\rightleftharpoons}} \underset{\substack{H^+\text{ acceptor}\\(\text{base})}}{HCO_3^-} + \underset{\substack{\text{Hydrogen}\\\text{ion}}}{H^+}$$

The presence of substances called buffers allows biological fluids to maintain a relatively constant pH despite the addition of acids or bases. A **buffer** is a substance that minimizes changes in the concentrations of H^+ and OH^- in a solution. It does so by accepting hydrogen ions from the solution when they are in excess and donating hydrogen ions to the solution when they have been depleted. Most buffer solutions contain a weak acid and its corresponding base, which combine reversibly with hydrogen ions.

39) One of the buffers that contribute to pH stability in human blood is carbonic acid (H_2CO_3). Carbonic acid is a weak acid that, when placed in an aqueous solution, dissociates into a bicarbonate ion (HCO_3^-) and a hydrogen ion (H^+). (See figure.)

$$\underset{\text{Carbonic acid}}{H_2CO_3} \rightleftharpoons \underset{\text{Bicarbonate ion}}{HCO_3^-} + \underset{\text{Hydrogen ion}}{H^+}$$

If the pH of blood drops, one would expect _____.

A) a decrease in the concentration of H_2CO_3 and an increase in the concentration of HCO_3^-

B) the concentration of bicarbonate ions (HCO_3^-) to increase

C) the HCO_3^- to act as a base and remove excess H^+ by the formation of H_2CO_3

D) the HCO_3^- to act as an acid and remove excess H^+ by the formation of H_2CO_3

Question-1:

(b) Calculate the surface-to-volume ratio for the spherical bacterium *Neisseria gonorrhoeae* (diameter 0.5 μm), responsible for the disease gonorrhea. Compare it with the surface-to-volume ratio for a globular amoeba, a large eukaryotic cell (diameter 150 μm). The surface area of a sphere is $4\pi r^2$.

Question-2:

(a) Calculate the axial length of an α-helix containing 78 amino acids. (b) How long would the polypeptide chain be if it were fully extended?

Question-3

6. Amino Acid Sequence and Protein Structure Our growing understanding of how proteins fold allows researchers to make predictions about protein structure based on primary amino acid sequence data. Consider the following amino acid sequence.

Ile	Ala	His	Thr	Tyr	Gly	Pro	Phe	Glu	Ala	Ala	Met	Cys	Lys	Trp	Glu	Ala	Gln	Pro	Asp	Gly	Met	Glu	Cys	Ala	Phe	His	Arg
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28

(a) Where might bends or β turns occur?

(b) Where might intrachain disulfide cross-linkages be formed?

(c) Assuming that this sequence is part of a larger globular protein, indicate the probable location (the external surface or interior of the protein) of the following amino acid residues: Asp, Ile, Thr, Ala, Gln, Lys. Explain your reasoning. (Hint: See the hydropathy index in Table 3–1.)

Question-4/solution-4

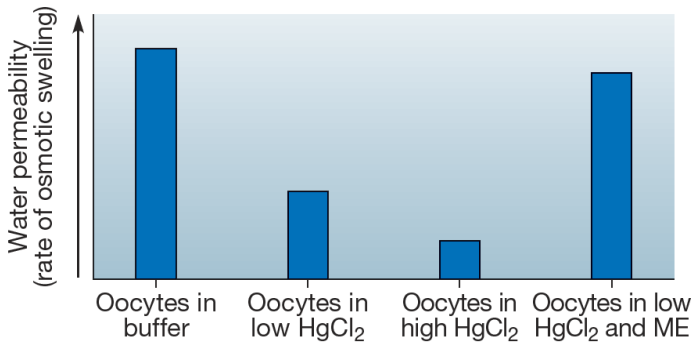
1. Relationship between Affinity and Dissociation Constant Protein A has a binding site for ligand X with a K_d of 10^{-6} M. Protein B has a binding site for ligand X with a K_d of 10^{-9} M. Which protein has a higher affinity for ligand X? Explain your reasoning. Convert the K_d to K_a for both proteins.

Level 1: Knowledge/Comprehension

3. Which best describes the structure of a cell membrane?
   a. proteins between two bilayers of phospholipids
   b. proteins embedded in a bilayer of phospholipids
   c. a bilayer of protein coating a layer of phospholipids
   d. cholesterol embedded in a bilayer of phospholipids
4. A plant cell placed in distilled water will _____; an animal cell placed in distilled water will _____.
   a. burst ... burst
   b. become flaccid . . . shrivel
   c. become turgid . . . be normal in shape
   d. become turgid . . . burst
5. The sodium concentration in a cell is 10 times less than the concentration in the surrounding fluid. How can the cell move sodium out of the cell? (*Explain your answer.*)
   a. passive transport
   b. receptor-mediated endocytosis
   c. active transport
   d. facilitated diffusion
6. The synthesis of ATP from ADP and P
   a. stores energy in a form that can drive cellular work.
   b. involves the hydrolysis of a phosphate bond.
   c. transfers a phosphate, priming a protein to do work.
   d. is an exergonic process.
7. Facilitated diffusion across a membrane requires _____ and moves a solute _____ its concentration gradient.
   a. transport proteins . . . up (against)
   b. transport proteins . . . down
   c. energy and transport proteins . . . up
   d. energy and transport proteins . . . down
8. What are the main types of cellular work? How does ATP provide the energy for this work?

Question-6:

Mercury is known to inhibit the permeability of water channels. To help establish that the protein isolated by Agre's group was a water channel (see Module 5.7), the researchers incubated groups of RNA-injected oocytes (which thus made aquaporin proteins) in four different solutions: plain buffer, low concentration and high concentration of a mercury chloride (HgCl2) solution, and low concentration of a mercury solution followed by an agent (ME) known to reverse the effects of mercury. The water permeability of the cells was determined by the rate of their osmotic swelling. Interpret the results of this experiment, which are presented in the graph below.



Data from G. M. Preston et al., Appearance of water channels in *Xenopus* oocytes expressing red cell CHIP28 protein, *Science* 256: 3385–7 (1992).

Control oocytes not injected with aquaporin RNA were also incubated with buffer and the two concentrations of mercury. Predict what the results of these treatments would be.